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Functional load modulates acoustic and articulatory variability in English

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Information-theoretic conditioning of speech variation

- Different information-theoretic (Shannon 1948) correlates are shown to predict different types of variation:
- More informative/unpredictable items maintain higher distinctiveness/are more resilient to variation.
- For example:
  - Frequencies correlate w/ word durations (Baker & Bradlow 2009) & vowel merger rates (Kang, Yoon & Han 2015).
  - Functional load (# of minimal pairs) correlates w/ merger rates & vowel lengths (Wedel *et al.* 2013).

Shannon, C. E. (1948). A Mathematical Theory of Communication. *Bell System Technical Journal, 27*, 379–423. Baker, R. E., & Bradlow, A. R. (2009). Variability in word duration as a function of probability, speech style, and prosody. *Language and Speech, 52*(4), 391–413. Kang, Y ., Y oon, T. J., & Han, S. (2015). Frequency effects on the vowel length contrast merger in Seoul Korean. *Laboratory Phonology, 6*(3-4), 469–503. Wedel, A., Jackson, S., & Kaplan, A. (2013). Functional load and the lexicon: Evidence that syntactic category and frequency relationships in minimal lemma pairs predict the loss of phoneme contrasts in language change. *Language and Speech, 56*(3), 395–417.

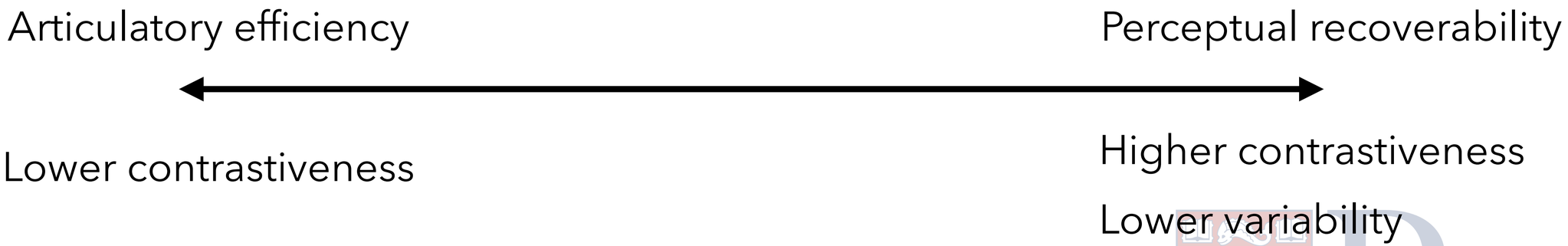
General findings: perceptual recoverability

- More informative items tend to resist variation more.
  - Frequent words have shorter durations (Baker & Bradlow 2009).
  - Higher functional load and lower word frequency lead to lower vowel merger rates (Wedel *et al.* 2013; Kang, Yoon & Han 2015).

Wedel, A., Jackson, S., & Kaplan, A. (2013). Functional load and the lexicon: Evidence that syntactic category and frequency relationships in minimal lemma pairs predict the loss of phoneme contrasts in language change. *Language and Speech, 56*(3), 395–417. Kang, Y ., Y oon, T. J., & Han, S. (2015). Frequency effects on the vowel length contrast merger in Seoul Korean. *Laboratory Phonology, 6*(3-4), 469–503.

Perceptual recoverability, articulatory efficiency & speech flexibility

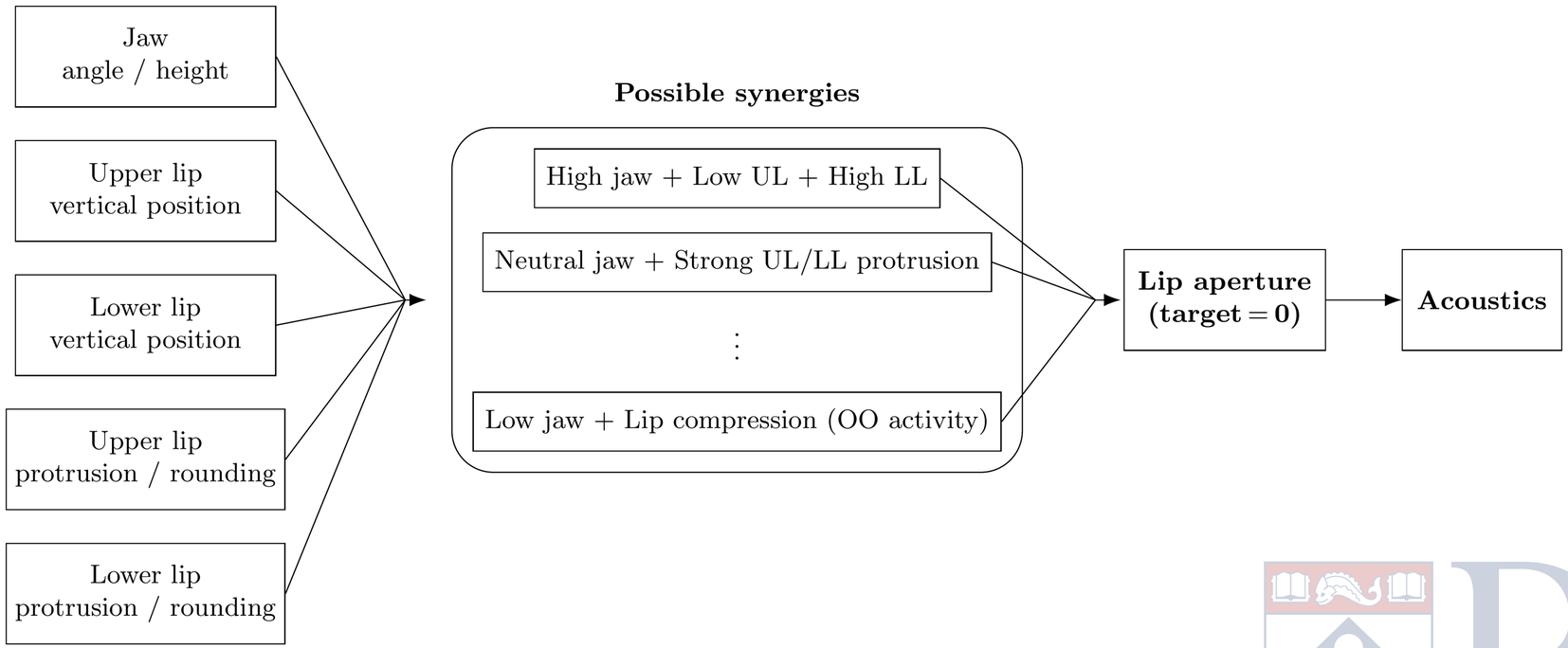
- Speech production is often regarded as conditioned by:
  - Perceptual recoverability pressures
  - Articulatory efficiency pressures (Jun, 1995; Kirchner, 2001; Flemming, 2004; Cohen Priva, 2017)



Jun, J. (1995). Perceptual and Articulatory Factors in Place Assimilation: An Optimality Theoretic Approach (PhD dissertation). University of California, Los Angeles. Cohen Priva, U. (2017). Informativity and the actuation of lenition. Language, 93(3), 569–597. Flemming, E. (2004). Contrast and perceptual distinctiveness. In B. Hayes, R. Kirchner, & D. Steriade (Eds.), Phonetically Based Phonology (pp. 232–276). Cambridge University Press.

Perceptual recoverability, articulatory efficiency & speech flexibility

- Speech flexibility (i.e., motor equivalence; cf. Perrier & Fuchs 2015) suggests that the same target can be achieved through an array of motor movements.



Perrier, P., and Fuchs, S. (2015) Motor Equivalence in Speech Production. In M. A. Redford (eds.), *The Handbook of Speech Production*, (225–247). New Jersey: John Wiley & Sons.

Perceptual recoverability, articulatory efficiency & speech flexibility

- Speech flexibility (i.e., motor equivalence; cf. Perrier & Fuchs 2015) suggests that the same target can be achieved through an array of motor movements.
- Certain degrees of freedom exist, allowing for an intended speech target to be achieved via an array of articulatory synergies.
  - Perceptual recoverability and articulatory efficiency could be acheived at the same time.
- Consequently, such flexibility may be a window for the speaker to react differently in the face of variable information-theoretic constraints in the articulatory & acoustic domains.

Perrier, P., and Fuchs, S. (2015) Motor Equivalence in Speech Production. In M. A. Redford (eds.), *The Handbook of Speech Production*, (225–247). New Jersey: John Wiley & Sons.

Research questions

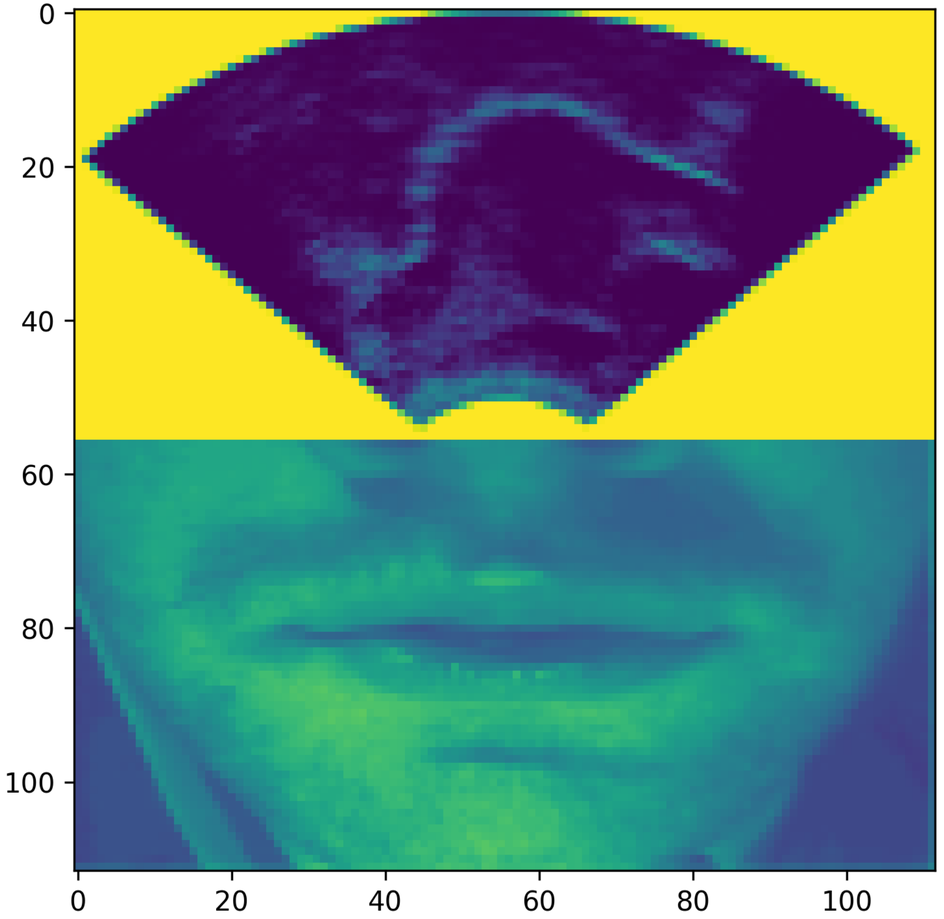
- Is the functional load effect generalizable across phonemic minimal pairs?
  - Do functionally loaded pairs have higher distinctiveness from each other?
- Does functional load show a positive effect in acoustics (perceptual recoverability) but a weaker or negative effect in articulation (articulatory efficiency)?
- Does speech flexibility enlarge this modality difference?

Methods

- Data
- Autoencoded speech representations
- Distinctiveness between two phonemes
- Functional load calculation
- Speech flexibility estimation

Data

- We collected ultrasound/lip imaging data from the Tongue and Lips Corpus (Ribeiro *et al*. 2021).
- Read and spontaneous speech produced by British English speakers



Ribeiro, M. S., Sanger, J., Zhang, J.-X., Eshky, A., Wrench, A., Richmond, K., & Renals, S. (2021) TaL: A synchronised multi-speaker corpus of ultrasound tongue imaging, audio, and lip videos. In Proceedings of the ieee workshop on spoken language technology. Shenzhen, China.

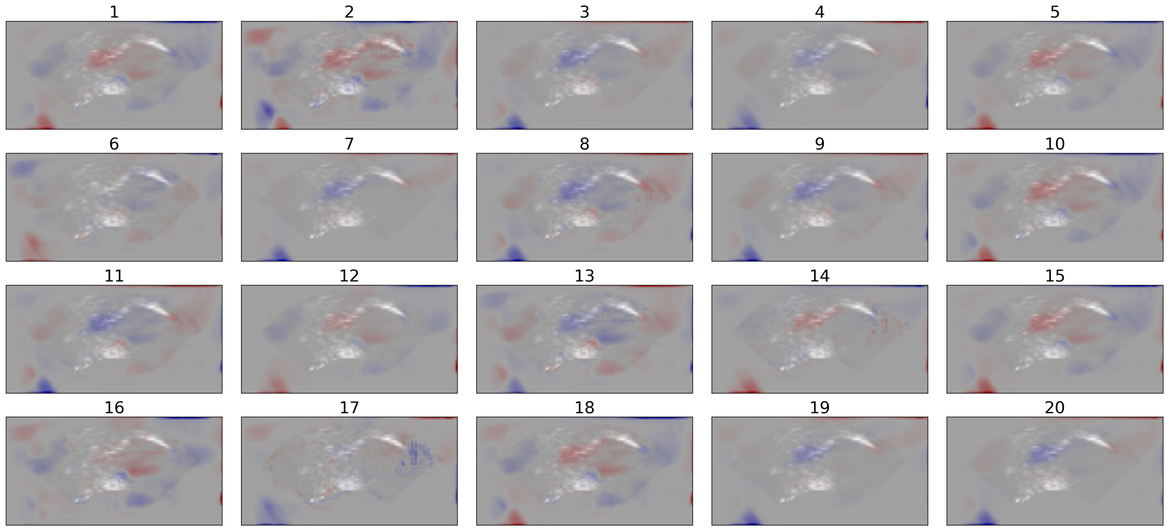
Autoencoded speech representations

- We investigated not just specific mergers (e.g., vowels) but all possible phonemic minimal pairs.
- This requires measuring different speech dimensions.
- We trained an autoencoder to encode raw acoustics/ultrasound-lip videos into latent speech representations (Huang & Breiss, 2026).
  - Acoustic/articulatory data → encoder → latent space (64D) → decoder → reconstructed data

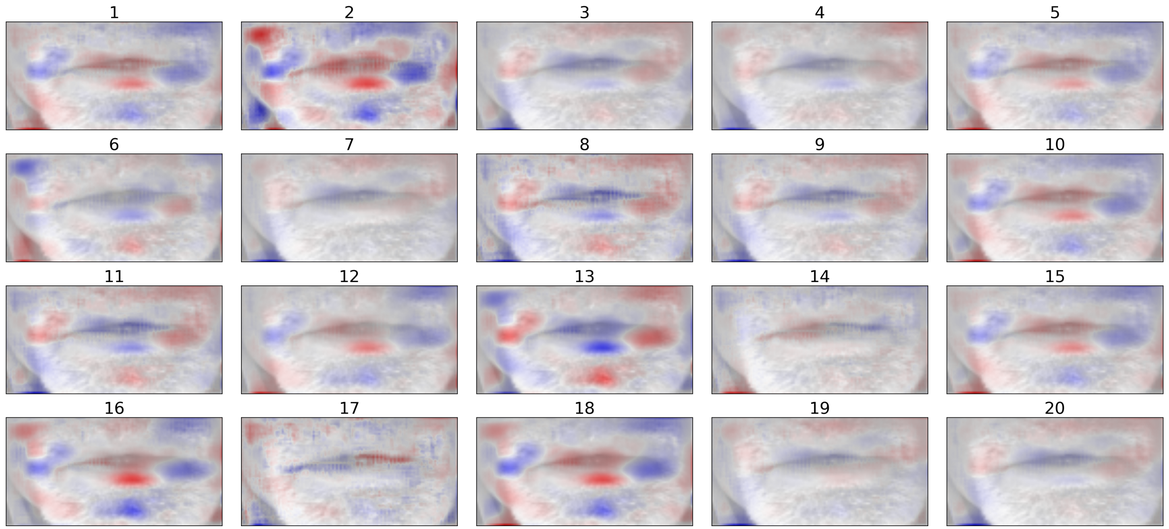
Huang, Po-Hsuan & Breiss, Canaan. (2026b, June 26–28). The conditioning of surprisal and many-to-one mappings in speech variation: Articulatory efficiency and perceptual recoverability. LabPhon 20, Montreal, Canada.

Autoencoded speech representations

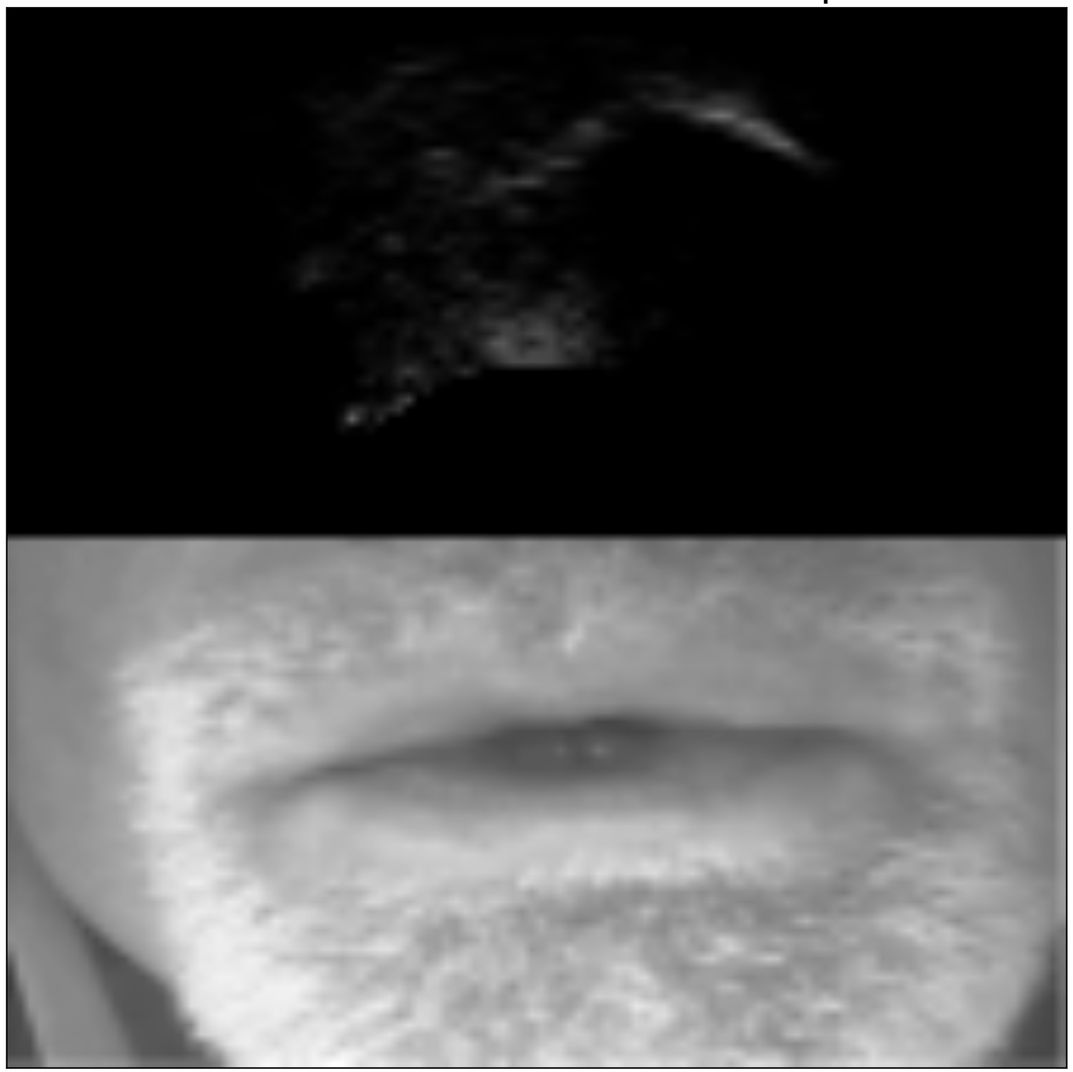
Ultrasound: latent dimension (/u/ in “lose”)



Lips: latent dimension (/u/ in “lose”)

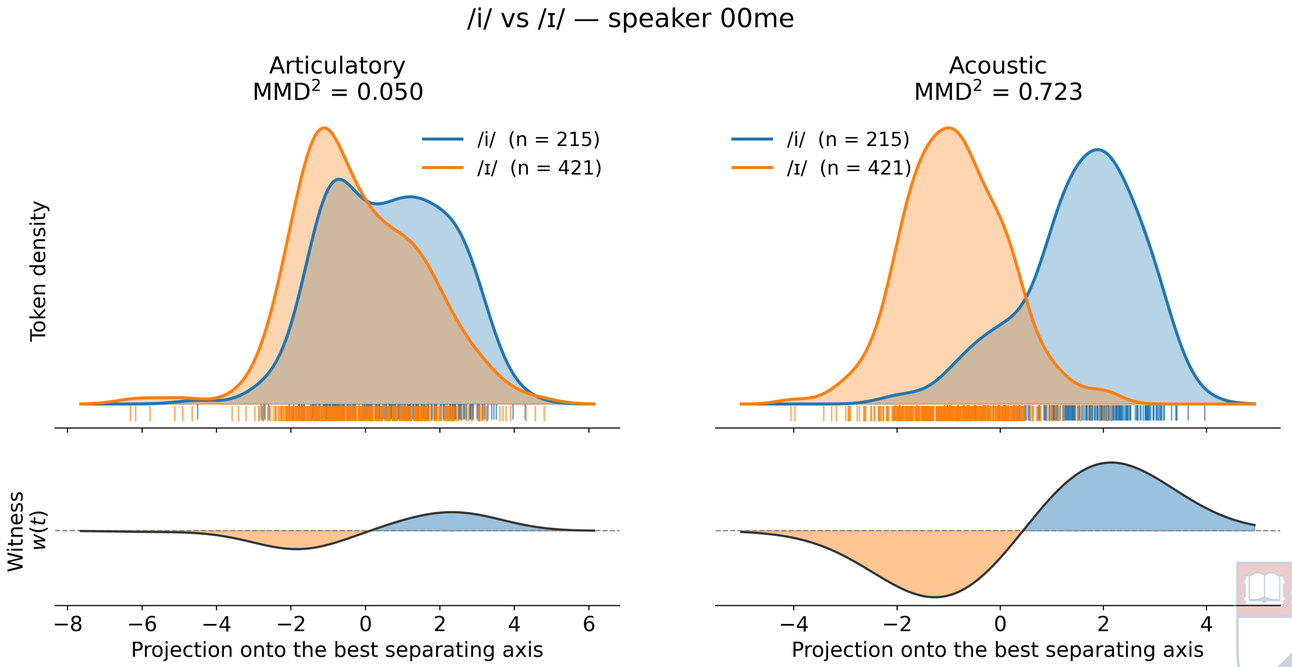


/u/ in “lose”: ultrasound over lips



Distinctiveness between two phonemes

- The acoustic/articulatory distinctiveness between two phonemes was calculated as the two phonemes' Maximum Mean Discrepancy in the latent spaces.



Unbiased MMD^2: zero means the two distributions are indistinguishable; small negative values are estimation noise.

Functional load calculation

- Lexical (token) frequency from the British National Corpus (BNC Consortium, 2007)
- Functional load = the entropy difference with and without the minimal pair

Dimension	Pairs
Vowel height	/i-ɪ/, /e-ɪ/, /ɛ-e/, /ɛ-a/, /ɑ-ɔ/, /ʊ-u/
Place of articulation	/n-ŋ/, /m-n/
Voicing	/k-g/, /b-p/, /t-d/, /ʃ-ʒ/, /s-z/, /f-v/, /ʧ-ʤ/
Liquids and glides	/l-ɹ/, /l-w/, /ɹ-w/

$$H = -\sum_{i \in W} p_i \log_2 p_i$$

$$H\Delta = H_1 - H_2$$

BNC Consortium. (2007). T*he British National Corpus, version 3* (BNC XML Edition). Oxford: Oxford University Computing Services.

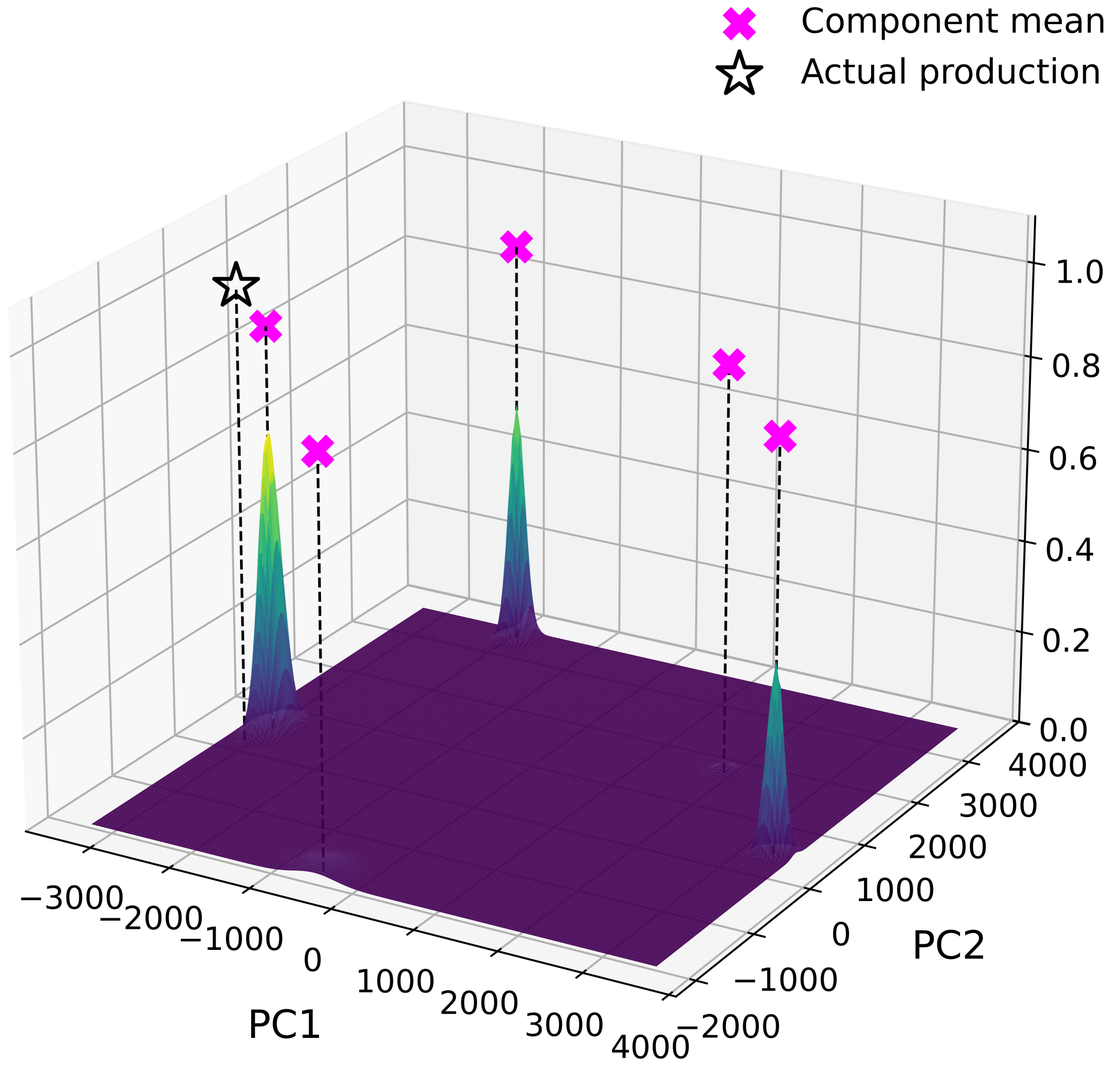
Speech flexibility

- Flexibility of each phoneme type was estimated as its conditional entropy through mixture density networks (MDN; cf. Richmond 2006; Zen & Senior 2014).
- MDN predicts a K-mixture (K=5) probability density distribution based on the input conditions.
- In this study:
  - P(observed articulatory matrix | acoustic matrix)

Richmond, K. (2006). A trajectory mixture density network for the acoustic-articulatory inversion mapping. In INTERSPEECH 2006 – ICSLP Ninth International Conference on Spoken Language Processing (Paper 1790-Mon3WeS.4). International Speech Communication Association. Zen, H., & Senior, A. (2014). Deep mixture density networks for acoustic modeling in statistical parametric speech synthesis. In Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP).

Speech flexibility

- A /l/ token produced by 00me (PCA used for illustration purposes).

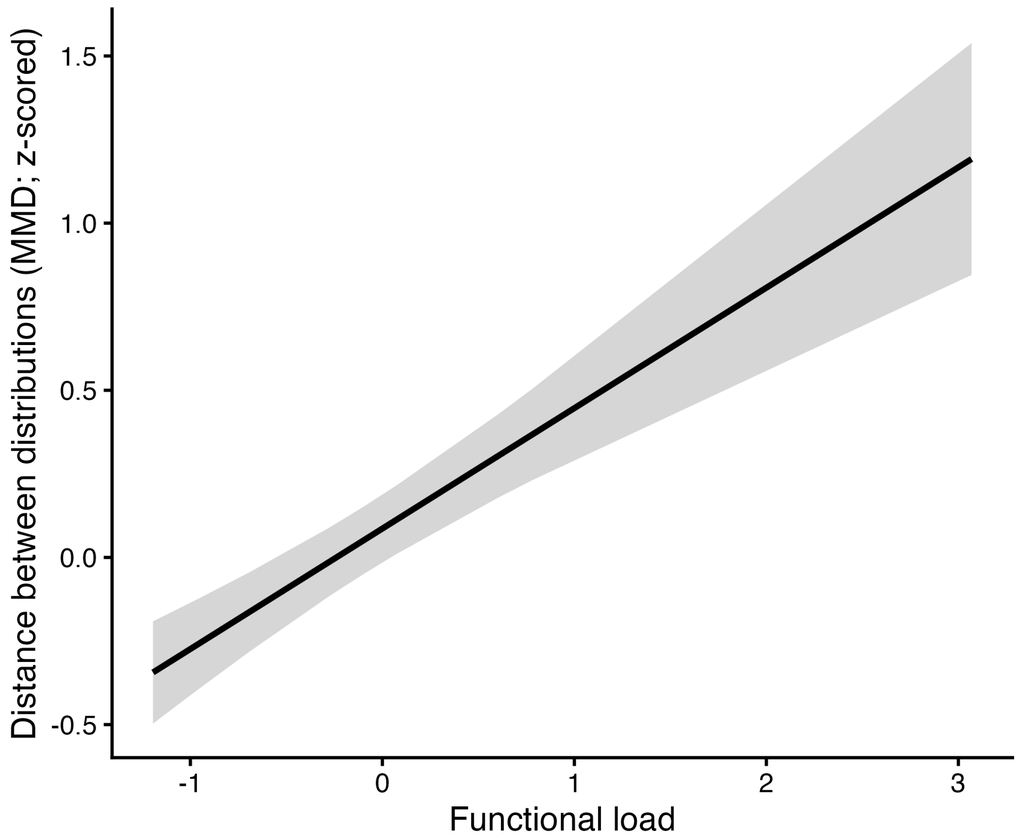


Statistical analysis

- Linear mixed-effects models were fitted to predict distinctiveness.
  - Predictors were surprisal, phone1/phone2 flexibility, distinctiveness type (articulatory vs. acoustic), phone2 frequency and phone2 duration.
  - Random intercepts/slopes on surprisal were grouped by speaker, word, and bigram.
- Bigrams that crossed syllable boundaries were excluded.

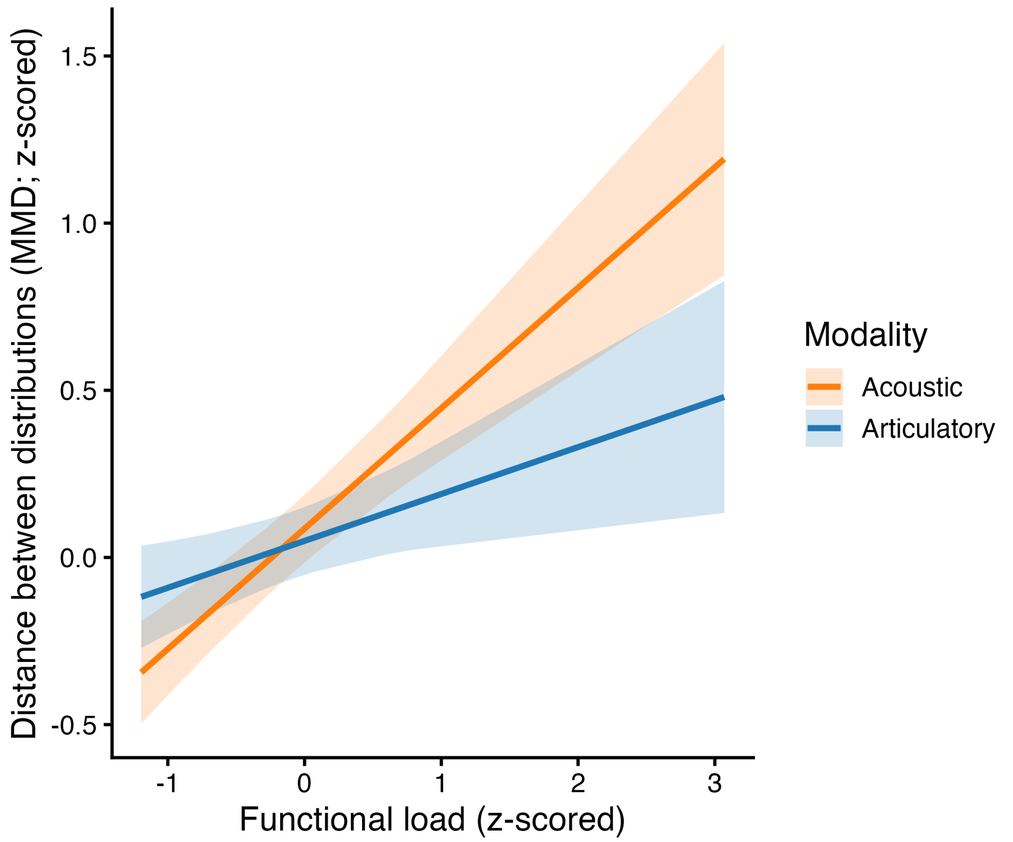
Results

- More functionally loaded pairs are further apart.



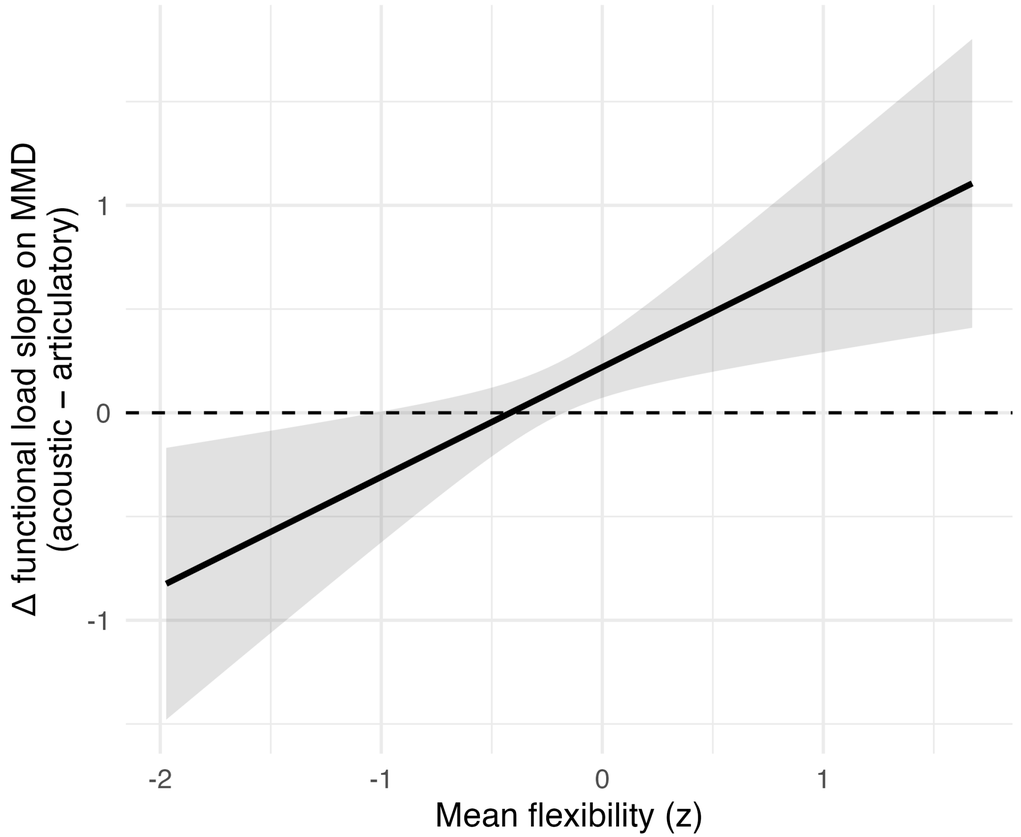
Results

- The functional load effect is weaker in articulation.



Results

- Flexibility:
  - weakens functional load effects in articulation.
  - enhances functional load effects in acoustics.



Discussion

- Functionally loaded phoneme pairs are more distinct from each other in speech.
- Such effects are weaker in articulation.
- When there is a stronger many-to-one mapping relationship:
  - the functional load effects are stronger in acoustics.
  - the functional load effects are weaker in articulation.

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Autoencoders